Bilingual retrieval augmented access to archived ANVISA regulatory documents

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Abbreviations: AI, artificial intelligence; ANVISA, Agência Nacional de Vigilância Sanitária; AREE, equivalent foreign regulatory authority; BM25, Best Matching 25; CI, confidence interval; CLS, classification token; DDCM, Clinical Drug Development Dossier; DPI, Dossiê do Produto sob Investigação; DSUR, Development Safety Update Report; FDA, U.S. Food and Drug Administration; GMP, good manufacturing practice; ICH, International Council for Harmonisation; MRR, mean reciprocal rank; RAG, retrieval-augmented generation; RRF, reciprocal rank fusion; SUSAR, suspected unexpected serious adverse reaction.

Abstract

Background. Bilingual regulatory review requires complete source-supported answers, not merely citations.

Objective. To evaluate retrieval-augmented generation (RAG) against archived Agência Nacional de Vigilância Sanitária (ANVISA) documents.

Methods. Six overlapping historical stages produced 404 outputs (150 internal, 75 English repair, 75 Portuguese repair, 50 stress, 24 calibration, 30 post-calibration). Coauthor review covered 94 historical outputs and a separate 40-answer assessment (20 bilingual pairs). An 80-answer repair comparison received author review. Further comparisons included 120 coauthor-rated outputs with hybrid passages, BM25 passages or no retrieval, and 32 positive-target retrieval questions.

Results. Historical stress completeness averaged 0.943 (40 outputs; 95% pair-bootstrap interval 0.873 to 0.990). The coauthor-reviewed assessment passed 10/20 pairs (exact 95% confidence interval [CI] 27.2 to 72.8%): 6/12 answerable, 0/4 partly answerable, 4/4 unanswerable. Author-reviewed repair success was 14/20 versus 17/20 (exact paired p = 0.375). Separate coauthor-rated success was 19/20 hybrid pairs (95% CI 75.1 to 99.9%), 18/20 BM25 pairs (68.3 to 98.8%) and 4/20 model-only pairs (5.7 to 43.7%). Hybrid minus BM25 success was +5 percentage points (95% pair-bootstrap interval -10 to +20). BM25 retrieved all required articles in 30/32 questions versus 32/32 for hybrid retrieval.

Conclusion. Retrieved evidence improved answers on these archived tasks, without establishing hybrid superiority over BM25. Citations did not ensure completeness or applicable law. Current-law accuracy and fresh independent validation remain unestablished.

Introduction

Regulatory document review begins with practical questions: which provisions and exceptions must a team check when planning a clinical-trial submission, preparing pharmacovigilance documentation or reviewing medical-device labelling? For example, a sponsor examining suspension of a Clinical Drug Development Dossier (DDCM) would need to locate the documentation, notification deadlines and implementation conditions in RDC 945/2024, Articles 85 and 89 [1]. Other review tasks include checking Development Safety Update Report (DSUR) submission requirements, adverse-event notification elements and conditions for non-printed device instructions. These examples motivate document review; they are not legal advice or evidence that this prototype answers current operational questions correctly. The suspension example uses RDC 945/2024 rather than Article 50 of the revoked RDC 9/2015 retained in some archived benchmark answers [1,2].

A bilingual regulatory team may ask in English while the relevant source is in Portuguese. The practical task extends beyond finding a related document: a reviewer needs the applicable article, its complete list of conditions and exceptions, an answer in the requested language, and a clear statement of what the selected evidence cannot establish. An omitted notification category could leave a draft checklist incomplete; a refusal that suppresses supported information could obscure a requirement that warrants review. These are plausible consequences motivating the evaluation, not observed effects on submissions or patient outcomes.

Retrieval-augmented generation (RAG) retrieves source passages before generating an answer [3,4]. A study of U.S. Food and Drug Administration (FDA) guidance search found that locating the correct source did not guarantee a complete and correct answer [5]. The gap addressed here is whether article-level traceability is accompanied by complete task performance when Portuguese regulatory text must support answers in two languages, including questions whose requested details are only partly available. Retrieval accuracy, citation support, response language, evidence boundaries and legal applicability therefore need to be examined separately.

This study evaluated that gap in a prototype indexing six archived ANVISA documents. We asked whether it could retrieve the cited provision, preserve the requested requirements and exceptions, answer in the question's language, and distinguish supported information from unavailable detail. The evaluation also examined whether apparent repair gains survived question-overlap audits, selected human review and checks of reference scope. The study was designed to assess traceable access to a fixed document set; it did not evaluate regulatory advice, live legal monitoring or decisions in an operational workflow.

The intended practical contribution is evidence for designing and evaluating assisted document review. Our experiments identify failure modes and evaluation checks for such a workflow. They do not demonstrate faster review, fewer submission errors or readiness for autonomous use.

Materials and Methods

Study design. This was a technical evaluation of a closed-corpus bilingual RAG prototype. The corpus was closed because the system was not allowed to search current web pages, submission portals, fee schedules, private company information, unpublished documents, or documents outside the local index. This design made the evidence boundary explicit. Unavailable detail required an explicit limitation; the revised partial-answer endpoint also required any supported information. The later model-only ablation supplied no passages or web-search tool and permitted stored knowledge under a common prompt.

The evaluation was developed in stages. Initial tests identified problems in corpus cleaning, retrieval, English questions over Portuguese documents, answer completeness, citation behavior, and refusal behavior. Repairs were made on development data. The final reporting separates development runs, internal baseline and repair runs, the first AI-authored diagnostic stress test, and a later refusal-focused AI-authored test.

Regulatory corpus. The corpus included six selected ANVISA documents covering clinical trials, biological products, medical devices, pharmacovigilance, and reliance-based review [1,2,6-9]. The selected documents concern common drug-development and health-product topics; this selection did not establish that every indexed provision was in force at evaluation. The final corpus included IN 338/2024, RDC 406/2020, RDC 55/2010, RDC 751/2022, RDC 945/2024, and RDC 9/2015. Corpus composition and use cases are summarized in Table 1. RDC 9/2015 was already revoked by Article 98 of RDC 945/2024 when these evaluations were conducted; Anvisa identifies 1 January 2025 as the latter regulation's effective date. Its presence in the archived corpus and gold answers is a benchmark limitation, not a deliberately designed temporal-validity test. Article 90 of RDC 945/2024 contains submission-date transitional provisions; applicability cannot be inferred for a benchmark question without the relevant dates [1,2].

Each source file was checked against its document identifier and topic before final indexing. Source verification identified one local-file metadata mismatch: the local file name suggested stability studies, while the document content was RDC 406/2020 on pharmacovigilance. The final corpus table therefore used official regulation numbers and topics rather than local file names. Identifier and topic checks did not establish consolidated legal validity. A subsequent status audit checked official texts on 23 October 2026 and identified 18 additional constructed-benchmark rows citing wholly revoked articles of RDC 55/2010, alongside the 28 rows citing revoked RDC 9/2015 (46 of 200 rows, or 23 bilingual pairs). The RDC 55/2010 revocations follow RDC 875/2024 and RDC 948/2024 [10,11]. Consolidated texts also show amendments under IN 345/2025, RDC 777/2023 and RDC 967/2025 for IN 338/2024, RDC 751/2022 and RDC 406/2020, respectively. Supplementary Section S4 records the source checks and their limits; whole-article exposure counts do not certify the remaining gold answers.

System and retrieval

The Python implementation used LlamaIndex, sentence-transformers, BM25 and Streamlit [12-17]. Text cleaning retained document, chapter, section, article and citation metadata. Articles up to 500 cl100k_base tokens remained intact; longer articles were split with a 400-token target and 60-token overlap before heading context was added. Embeddings used all-MiniLM-L6-v2 with the recorded CLS-pooling wrapper and a 512-WordPiece limit. BM25 and dense retrieval each returned 100 candidates; reciprocal-rank fusion used $k = 60$. A multilingual cross-encoder reranked candidates, and the top five passages supplied answer context. Figure 1 summarizes the workflow; Supplementary Section S1 records settings, prompts and scoring definitions. Implementation files, chunking settings, index audit and retrieval settings are detailed in Supplementary Tables S1, S2, S3 and S4, respectively.

English queries used a deterministic 114-entry Portuguese glossary and question-specific query, document and article anchors. Conditional neighboring-article and sibling-chunk expansion added context before generation. These rules were developed with exposure to benchmark failures; their evaluation uses development data. Article identifiers do not establish that every required clause reached the generator. The original prompt required a same-language, article-cited answer from retrieved context and a fixed refusal when evidence was insufficient. Glossary examples and fixed refusal sentences are listed in Supplementary Tables S5 and S6.

Historical generation and judging both requested the unpinned alias models/gemini-flash-lite-latest. Resolved July backend versions were not preserved and cannot be inferred from current documentation [17]. Later experimental responses recorded gemini-3.5-flash-lite. Historical timestamps, merge provenance and the distinction between chunking and embedding token units are documented in Supplementary Sections S1 and S5.

Assessment sets and overlap

The constructed benchmark contained 200 question rows in 100 bilingual pairs: 140 direct-article, 20 multi-article/exception and 40 unanswerable/adversarial questions (Figure 2). This construction total differs from the evaluated denominators. The internal assessment used 150 questions; each language-specific repair reused its 75-question internal subset with identical IDs and text. The other

stages comprised 50 stress, 24 calibration and 30 post-calibration outputs. Together the six stages produced 404 outputs and 254 distinct question IDs, without semantic independence of those IDs (Supplementary Section S2). Constructed benchmark composition is detailed in Supplementary Table S7.

The stress findings informed refusal calibration. Three substantively equivalent bilingual pairs recurred between stress and calibration, covering the current SUSAR form, device-notification fee and patient-identification premise; four further pairs had related content with different framing. The repair, calibration and post-calibration results therefore describe development and cannot establish independent held-out performance or a causal calibration effect.

Historical scoring and coauthor review

Automated metrics assessed evidence support, citation correctness, completeness and refusal, using continuous scores and archived labels [12,18,19]. Historical confidence intervals resampled bilingual pair IDs together with 2,000 percentile-bootstrap draws (seed 20260619). They are conditional on the archived labels, sources and judge and exclude uncertainty from legal-status or labeling errors. The same model family generated and judged answers, limiting independence. Audit flags, bootstrap settings, stage denominators and historical estimates are given in Supplementary Tables S8, S9, S10 and S11, respectively.

On 26 October 2026, H.L.R.J. reviewed and approved 94 selected historical outputs, including all 44 originally unanswerable stress/calibration/post-calibration outputs, 40 originally answerable outputs and ten unanswerable internal outputs. The ratings began as an AI-prepared draft; the coauthor had seen some scores, references or coordinator information. The review was author-led and unblinded. Completeness agreement used a judge threshold of 0.80, with 1.00 as sensitivity; refusal comparisons used the coauthor-defined unanswerable stratum. Original ratings and subsequent answerability adjudications were retained separately (Supplementary Section S3). Review sampling and author/judge agreement are reported in Supplementary Tables S12 and S13.

Frozen 40-answer author-led assessment

Twenty bilingual pairs (40 questions: 12 answerable, four partly answerable and four unanswerable pairs), references and scoring rules were frozen before generation on 30 October 2026. The original six-document index and prompt produced one answer per question with gemini-3.5-flash-lite at temperature 0.2, without retries, repairs or new automated judging. H.L.R.J. rated every answer against the exact selected passages and frozen reference on 30 October, reporting no exposure to automated scores for this run. Pair success required both languages to meet the category rules and requested language. Partly answerable questions required supported information and identification of the unavailable detail; unanswerable items had no supported-content completeness endpoint. Exact 95% binomial intervals used pairs. The AI-authored, coauthor-approved questions followed earlier benchmark exposure; this is author-led assessment, not independent validation (Supplementary Section S3).

Development diagnostics and exploratory repair

Retrospective retrieval ablations used 160 answerable development questions in 80 bilingual pairs. They compared BM25, dense retrieval, fusion, reranking and rule removal on the fixed historical index. A window-pooling comparison changed only the embeddings of 35 overlength nodes; a targeted answer experiment used 30 affected questions and 20 seeded controls, three repeats per condition, totaling 300 generations and 300 same-model automated evaluations. Paired differences averaged repeats before resampling pairs with 10,000 draws. These reused-question analyses did not select a deployment setting (Supplementary Sections S5 and S6). Retrieval conditions and paired contrasts are specified in Supplementary Tables S14 and S15.

Following the 40-answer assessment, controlled development experiments varied source completeness, instructions, source headings, language cues and reference scope. Negative controls and failed repairs were retained. The coauthor separately reviewed 44 first-repeat sibling-context answers after exposure to scores and conditions. These experiments diagnose selected failures and

do not create new independent test items (Supplementary Sections S6 and S7). Sibling-context outcomes, mechanism controls and archived-reference decisions are summarized in Supplementary Tables S16, S17 and S18, respectively.

A separate 482-node index and bundled repair were frozen before question authoring. The repair combined structural parsing corrections, source-occurrence article completion, English-to-Portuguese retrieval translation, a named-regulation filter, language cues, revised partial-answer instructions, an exact-subject evidence check and deterministic citation formatting. Twenty AI-authored bilingual pairs comprised 12 answerable, four partly answerable and four unanswerable pairs; screening against 452 historical questions replaced one semantic duplicate before freeze. Execution completed 20 translations and 80 answers by 1 November 2026 with gemini-3.5-flash-lite at temperature 0.2, without judge or repair calls. Pair success required content, evidence-boundary, language and citation-label criteria. Raw-output and common-formatter sensitivities were retained. All 80 outputs received personal author review on 1 November: 44 unblinded corresponding-author confirmations and 36 coauthor-reviewed AI-prepared ratings. The corresponding author reported that H.L.R.J. saw no separate scores, conditions or prior findings before finalizing those 36. Initial AI scores and author overlays remain separate (Supplementary Section S8).

Pair success required both languages to satisfy the specified criteria. Exact binomial intervals used pairs. The repair comparison also used paired bootstrap differences and an exact discordant-pair test. All intervals are conditional on the selected questions and rubric and omit uncertainty from source dependence, review exposure and AI-draft assistance.

Post hoc common prompt and lexical baseline comparisons

On 2 November 2026, the existing 40 FV questions were used for 80 new answers with gemini-3.5-flash-lite at temperature 0.2. Identical neutral instructions permitted stored knowledge without passages; only the evidence block differed. RAG used saved integrated-repair contexts. Neither condition used web search. Expanded FV_012 scope included archived Articles 17, 30 and 31 before generation. Content and evidence-boundary success required complete supported information, no material unsupported claim, appropriate handling of unavailable detail and the requested language; appropriate abstention passed absent-detail items. Citations were secondary. Codex assessed anonymized outputs with limited masking. On 2 November, H.L.R.J. personally drafted and approved ratings for all 118 distinct answers representing 120 condition records, including the subsequent BM25 extension. He reported no prior AI scores, condition identities or findings for these outputs, but had project/question exposure and answer style could reveal conditions. Ratings used common references rather than generator contexts. AI and author ratings remain separate. Whole-pair bootstrap intervals used 20,000 draws (seed 20261102).

The separate lexical baseline used the same 482 chunks, saved Portuguese retrieval queries, named-regulation filter and five seed chunks as the frozen hybrid/reranked retriever. BM25 used $k1 = 1.5$, $b = 0.75$ and epsilon = 0.25, with no dense branch or reranker. Thirty-two answerable or partly answerable language items had positive gold article targets; eight absent-detail items were excluded from recall. Both methods used the same full-article and adjacent-article completion policy, with variable final context lengths (Supplementary Section S10).

Results

Historical development and diagnostic performance

The six-document index contained 382 parsed source records, including six preambles and repeated labels, and 481 chunks; 359 records remained intact and 23 were split. The benchmark covered 78 unique gold article citations. RDC 9/2015 appeared in 28/200 question rows. An additional 18 rows cited wholly revoked RDC 55/2010 articles, giving 46/200 exposed rows in 23 bilingual pairs. These are exposure counts, not adjudicated current-law error rates (Table 1; Supplementary Section S4).

Historical internal completeness averaged 0.808 across 120 originally answerable outputs. English completeness rose from 0.645 to 0.995 and Portuguese completeness from 0.970 to 0.995 on their reused repair subsets. The stress assessment instead showed completeness 0.943 (95% pair-bootstrap interval 0.873 to 0.990; 40 answerable outputs), with refusal correctness 7/10 (0.700; interval 0.300 to 1.000; five bilingual pairs) under the original judge and labels. Later post-calibration refusal correctness was 17/18 (interval 0.833 to 1.000; nine pairs), with answerable-control completeness 0.908 (interval 0.792 to 1.000; 12 outputs). Different, development-linked sets preclude a causal before/after claim. Original scores and stage denominators are retained in Supplementary Data S1 and S2. Historical estimates and intervals are listed in Supplementary Table S11.

Human review and answerability adjudication

At the 0.80 judge threshold, coauthor and automated completeness classifications agreed on 38/58 applicable outputs (65.5%; exploratory pair-bootstrap interval 48.2 to 80.9%): 37 both passed, one both failed, 16 passed the judge but failed author review, and four showed the reverse. Among the 46 coauthor-classified answerable outputs, agreement was 37/46 (80.4%; 63.8 to 93.6%). Citation-support agreement was 44/44, an all-pass result that did not discriminate quality. These targeted, unblinded agreement statistics do not estimate independent validation accuracy.

Archived-source adjudication reclassified six originally unanswerable outputs as answerable and twelve as partly answerable, preserving the original labels. Refusal of an unavailable detail was evaluated separately from completeness of supported information. In the retained stress, calibration and post-calibration unanswerable strata, author-rated appropriate refusals were 4/4, 10/10 and 12/12 outputs, or 2/2, 5/5 and 6/6 bilingual pairs. Exact pair-level 95% intervals were 15.8 to 100.0%, 47.8 to 100.0% and 54.1 to 100.0%. The twelve partly answerable outputs remained separate. These small, differently composed strata do not establish improvement (Supplementary Section S3).

Frozen coauthor-reviewed answers

In the frozen author-led assessment, 10/20 bilingual pairs met the full task criteria (50.0%; exact 95% CI 27.2 to 72.8%). Category results were 6/12 answerable pairs (50.0%; 21.1 to 78.9%), 0/4 partly answerable pairs (0%; 0 to 60.2%) and 4/4 unanswerable pairs (100%; 39.8 to 100%). At output level, 24/40 passed: 16/24 answerable, 0/8 partly answerable and 8/8 unanswerable. Full reference content was covered in 18/24 answerable outputs, but English language failures reduced complete-task success. Requested language was respected in 35/40 outputs: 15/20 English and 20/20 Portuguese. Table 2 and Figure 3 report the paired outcomes; Supplementary Section S3 provides all ratings and derived results.

The selected passages confirm missing list items in AV_004, AV_006 and AV_014, absence of the required Article 28 item in AV_013 Portuguese, and missing portions of the paired provisions in AV_016. AV_009 Portuguese correctly supplied the registration classes but could not supply the notification classes from its passages. All eight partly answerable outputs were blanket refusals. Five omitted information that the coauthor judged supported by the selected passages; three lacked the required supported content in those passages. None identified the unavailable detail explicitly. The frozen prompt required a blanket refusal for partial evidence, whereas the prespecified partly answerable endpoint required a supported partial answer. This instruction and endpoint mismatch is a documented design limitation. AV_008 English cited supported pharmacovigilance statements about risk minimization instead of answering the contingency-plan question. All 23 claim-bearing outputs had supported claims and citations according to the coauthor, illustrating why grounding alone is not task success. These failure categories overlap and do not identify causal effect sizes.

Retrieval and diagnostic findings

On reused development questions, article Recall@5 was 0.6969 for citation-diversified BM25, 0.3344 for dense retrieval, 0.5687 for fusion and 0.9500 after reranking without anchors or neighbors. The full pipeline reached 0.9875. BM25 candidates with the full downstream rules also reached 0.9875, concealing one gain and one loss when dense candidates were added; the paired interval for that difference was -0.0187 to +0.0187. Thus, the comparison did not demonstrate an incremental mean

benefit of the dense branch. These are retrieval endpoints, not answer-quality outcomes (Supplementary Section S5). Retrieval conditions and paired ablation contrasts are reported in Supplementary Tables S14 and S15.

The 300-answer window-pooling experiment changed affected-stratum automated completeness from 0.590 to 0.617 (paired difference +0.027; 95% interval -0.027 to +0.099). Improvement remained unproven. Scores also differed for byte-identical prompts, and all 300 automated faithfulness scores were 1.000, including 89 refusals. Selected passage completeness, source structure, language instructions and question-reference scope each contributed to case-level failures; increasing context alone did not eliminate refusal of supported content. Detailed controls, negative results and exposure limits are in Supplementary Sections S6 and S7.

Exploratory repair on frozen author-led questions

Completed personal author review passed 14/20 baseline pairs (70.0%; exact 95% CI 45.7 to 88.1%) and 17/20 repaired pairs (85.0%; 62.1 to 96.8%). Four pairs improved and one worsened: difference 15.0 percentage points (pair-bootstrap 95% interval -5.0 to +35.0), exact two-sided p = 0.375. The coauthor failed repaired FV_012 English (IRH_051) because the technical-report requirement is cited to Article 17 rather than Article 31 II. Expanded content coverage and citation support are distinct. Initial AI scores remain 14/20 versus 18/20 (difference 20.0 points; interval +5.0 to +40.0; p = 0.125). Initial AI raw-output and common-formatter results remain 14/20 versus 15/20 and 15/20 versus 18/20; these are historical sensitivities, not recomputed human endpoints (Table 3; Figure 4; Supplementary Section S8). Supplementary Table S19 separates initial AI assessments from completed author review. The four gains comprised two answerable and two partly answerable pairs; the citation-support regression occurred in one answerable pair. Thirteen pairs passed both conditions and two failed both, so the net change was three successful pairs.

Author-reviewed category results were 10/12 versus 11/12 answerable pairs, 0/4 versus 2/4 partly answerable pairs and 4/4 unanswerable pairs in both conditions. Staffing and ruminant-documentation answers still refused supported information despite complete reference passages. Both conditions selected complete frozen-reference wording in all 32 positive language rows, so no retrieval-coverage gain was demonstrated. Mean context length increased from 5,737 to 16,487 characters. AI-draft assistance and differing review exposure limit inference. Supplementary Table S20 separates initial AI-assessed category counts from completed author review. The repaired answerable count is 11/12 under author review, compared with the initial AI-assessed 12/12; the partly answerable and unanswerable counts are unchanged.

The corresponding author confirmed that broad FV_012 manufacturer/site and GMP scope requires archived Articles 17, 30 VIII-IX and 31 I-III of RDC 55/2010. Only repaired English supplied the expanded content, but it failed the specific citation above. Expanded-scope author review yields 13/20 versus 17/20 pairs. This output-informed sensitivity preserves original references and AI scores. Article 30 is revoked; the adjudication concerns the archived question (Supplementary Sections S4 and S8).

Post hoc common prompt and lexical baseline results

Coauthor-rated content and evidence-boundary success was 39/40 with hybrid passages, 38/40 with BM25 passages and 9/40 for the model alone, including eight appropriate abstentions per condition. Bilingual success was 19/20, 18/20 and 4/20 pairs (exact 95% intervals 75.1 to 99.9%, 68.3 to 98.8% and 5.7 to 43.7%). Hybrid minus model-only success was +75 percentage points (descriptive pair-bootstrap 95% interval +55 to +90; 15 gains, no losses; exact paired p = 0.000061); hybrid minus BM25 was +5 points (-10 to +20; two gains, one loss; p = 1.000). This content endpoint differs from the earlier full-pipeline endpoint. Omissions affected 0/40, 1/40 and 31/40 answers; material claim flags affected 1/40, 1/40 and 14/40. Author-rated citation presence was 39/40, 38/40 and 0/40, and correction flags were 1/40, 2/40 and 32/40, without measured editing effort. Topic results are in Supplementary Table S21 and Supplementary Figure S1 (Table 4; Figure 5).

In the matched retrieval baseline, all required articles appeared within five chunks for BM25 30/32 versus hybrid 32/32; full required article text after completion was present for 31/32 versus 32/32. Mean article Recall@5 was 0.958 versus 1.000 and MRR@5 was 0.932 versus 0.964. Pair-bootstrap difference intervals included zero. BM25 original-English queries found all required articles for 5/16 English questions, compared with 15/16 using the saved Portuguese translations; this language sensitivity is separate from the matched primary comparison (Supplementary Table S22).

The BM25 extension used 39 new first-attempt calls and one exact-prompt cached answer, with saved hybrid comparators. Its English FV_016 omission coincided with missing Article 22 in context; its Portuguese FV_012 answer added intermediate-product manufacturers to the explicit name/address list. The hybrid failure equated DPI with an Investigator's Brochure. Human and AI primary labels agreed on 117/118 distinct answers; one accepted model-only answer changed the AI count from 8/40 to 9/40 but not pair success. Secondary disagreements remain separate (Supplementary Section S10).

Discussion

Traceable citations and high retrieval scores did not ensure task success. The coauthor-reviewed assessment passed 10/20 bilingual pairs despite support for claims and citations in all claim-bearing outputs (Table 2; Figure 3). Reviewers still need to check lists, exceptions, language and task relevance. Missing notification classes and truncated items could leave a draft checklist incomplete; the wrong-plan answer supplied supported information about the wrong task, and five English questions received Portuguese answers. These are implications of benchmark failures, not observed effects on regulatory work.

Partly answerable questions require both supported information and an explicit boundary around unavailable detail. The original blanket-refusal instruction conflicted with that endpoint. Revised instructions and complete evidence improved selected development outcomes but retained failures and occasional task or formatting regressions. In the author-reviewed repair comparison, staffing and ruminant-documentation answers still withheld supported content. Four task-level gains coexisted with one citation-support loss and two persistent partial-answer failures (Figure 4). The difference was +15.0 points, with an interval from -5.0 to +35.0; reliable improvement was not established. Both conditions already selected complete frozen-reference wording, so retrieval-coverage gains cannot explain this comparison. Refusal assessment must address both invented claims and suppression of supported information.

The common-prompt comparison supports supplying evidence over stored knowledge alone on these tasks (Table 4; Figure 5). Its expanded reference, new outputs and content endpoint prevent direct comparison with earlier repair scores. Appropriate abstention counts as success on absent-detail questions. BM25 nearly matched hybrid retrieval coverage and answer quality; the paired interval does not establish superiority or equivalence. Different failures matter: BM25 omitted evidence and extended a checklist, while hybrid substituted the wrong dossier type. Coauthor confirmation strengthens assessment but does not remove development exposure. A strong lexical baseline is therefore necessary before attributing gains to retrieval complexity.

Corpus validity is a separate limitation. A targeted consolidated-text check retained the staffing criterion in RDC 406/2020 Article 18, but confirmed RDC 55/2010 Articles 22 and 30 were revoked by RDC 948/2024 [7,8,11]. The ruminant-documentation target and FV_012 Article 30 components therefore remain historical reading targets. Separate transmissible spongiform encephalopathy documentation rules require product-specific applicability review. Current-law corpus rebuilding, reference validation and model reevaluation remain incomplete; archived-text success must not be interpreted as current regulatory accuracy (Supplementary Section S4).

Fresh independent validation remains absent. AI-authored questions, development exposure, post hoc analyses, source dependence and single generations limit inference. Historical generation and judging shared an unresolved alias; window-pooling scores used the generation model and lack

complete human verification. The new comparison has coauthor assessment with prior question exposure and limited condition masking; this is not independent external validation. BM25 generation was sequential and reused saved hybrid comparators. The English embedding model was not compared with a multilingual replacement [20,21]. Repeated generations do not create independent questions.

A candidate workflow would present the question, answer, supporting passages and unavailable details for professional verification against the official current text. The prototype does not maintain consolidated law or reliably warn of superseded provisions. This study evaluates model answers and retrieval, not superiority over web search or professional workflow benefit. No review time, correction effort, decision quality or operational safety was measured. Future workflow testing would require frozen current-law references and realistic tasks.

Conclusion

This archived-corpus study addresses the gap between finding a regulatory citation and delivering a complete bilingual answer to a document-review question. Human review exposed omitted requirements, wrong-language and wrong-task answers, and refusal of supported information. The contribution is an auditable set of failure cases and evaluation checks: preserve full provisions, verify scope and language, and distinguish unavailable detail from supported content. Coauthor-rated answers favored retrieved evidence over the model alone, but hybrid retrieval showed no established advantage over matched BM25 retrieval. Current-law accuracy and fresh independent validation remain unestablished.

Declarations

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Conflict of Interest: The authors declare no competing financial interests.

Author Contributions: C.V.M.-P. conceived and designed the study, developed the software pipeline, prepared the corpus, constructed the benchmark, performed the analyses, interpreted the results, and drafted the manuscript. H.L.R.J. contributed to the regulatory and pharmaceutical framing, interpretation of results, author review of questions and generated answers, and critical revision of the manuscript.

Data Availability: Code, curated data, frozen outputs, separate AI and author ratings, and figure sources are archived as v0.2.1 at https://doi.org/10.5281/zenodo.23925532 and https://github.com/camontefusco/ANVISA_RAG_publication_repository_20260720/releases/tag/v0.2.1. The accompanying package adds these verified citations without changing scientific results. Historical outputs and original reviews remain separate from derived analyses. Full source/index corpora, credentials and private coordination keys are excluded.

Ethics statement: The study used public regulatory documents and generated benchmark questions. Human assessment consisted of coauthor review of model outputs. No patient data, personal health data, animal experiments or clinical trials were involved.

AI assistance disclosure: OpenAI Codex (OpenAI) was used to assist software implementation and debugging, data and source-file audits, analysis-script preparation, manuscript drafting and revision, and preparation of documents and figures from inspectable project data. AI-generated benchmark questions, answers and judge outputs are described separately in Methods and the supplementary records. Historical assistant model versions and complete session-level usage metadata were not consistently recorded. The authors retain responsibility for the manuscript, software and interpretations; AI assistance does not constitute independent human verification. Human regulatory adjudication still pending during this revision is identified explicitly in the relevant records.

Clinical Trial Registration: Not applicable.

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